Open-Motion Research

User Manual

Product	Open-Motion Research blood-flow monitor
Application version	1.4.0
Document revision	July 2026 — draft
Audience	Research / study operators

> **DRAFT — not a controlled document.** This manual is a documentation preview generated from Open-Motion Research 1.4.0 running on live hardware. Open-Motion Research is intended for investigational use only.

Contents

1. About the system

Open-Motion is Openwater's optical blood-flow monitor. Two forehead **sensor modules** (up to 8 cameras each) and a **console** containing a near-infrared laser measure laser-speckle images through the skin and compute, at 40 samples per second per camera:

- **BFI — Blood Flow Index**: relative index of blood flow (white trace).
- **BVI — Blood Volume Index**: relative index of blood volume (blue trace).

Open-Motion Research (window title "Open-Motion Research", orange **BETA** badge in the header) is the investigational variant of the Open-Motion application. It keeps the same core workflow and adds full acquisition control. Laser safety is enforced by a hardware interlock inside the console independently of this software.

What Open-Motion Research adds

Everything in the
Open-Motion User Manual
applies here as well. On top of it, Open-Motion Research provides:

Addition	Summary	§
Per-camera plot grid	One live plot cell per active camera instead of two side-average plots.	3, 9
Scan Settings	Session label, per-sensor camera patterns, timed/continuous duration.	4
Check button	On-demand contact-quality check, decoupled from starting a scan.	5
Direct scan start	Start launches the scan immediately; run Check yourself beforehand when needed.	6
Plot display options	Mean/Contrast display mode, Y-axis autoscale, axis labels, BVI low-pass filter, extra plot bounds.	9, 10
Software updates	Update banner at launch and version status in Settings → About.	2, 10

2. Starting the application

Launch **Open-Motion Research**. Only one copy can run at a time (a dialog appears if another instance already holds the hardware). The app connects to the console and both sensors **automatically**; after connecting it initializes the sensors for a few seconds. Allow the system to settle for about two minutes after connecting before starting a scan — starting too early can fail with a "camera not ready" condition.

If a newer application release is available, an update banner slides in under the header: *"A new version is available: vX.Y.Z"* with an **Update** button (downloads and installs in place) and an ✕ to dismiss.

The connection badge states (grey `Disconnected` / green `Start` / yellow pending / red `Stop`) are the same as in the *Open-Motion User Manual*, §2.1.

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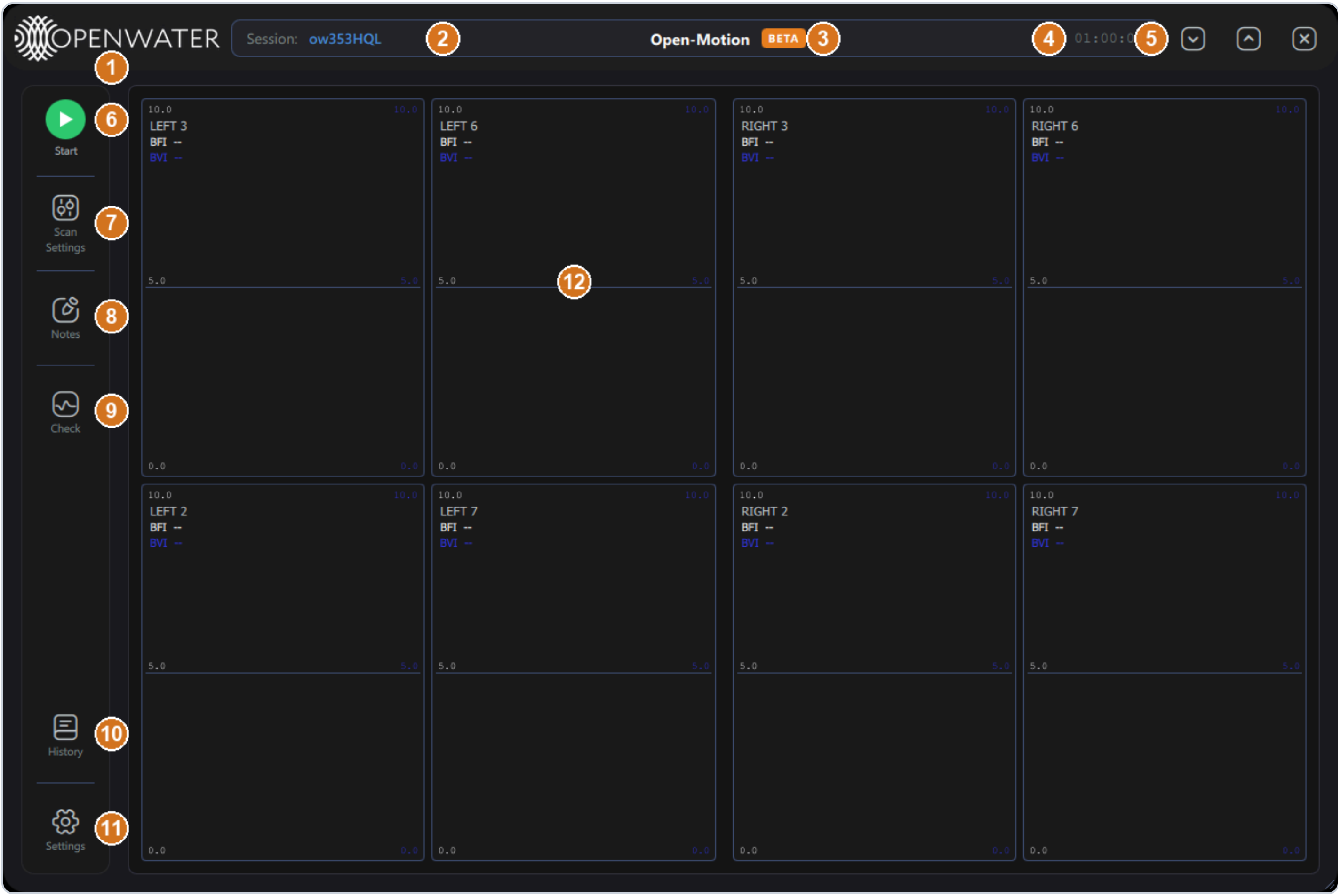
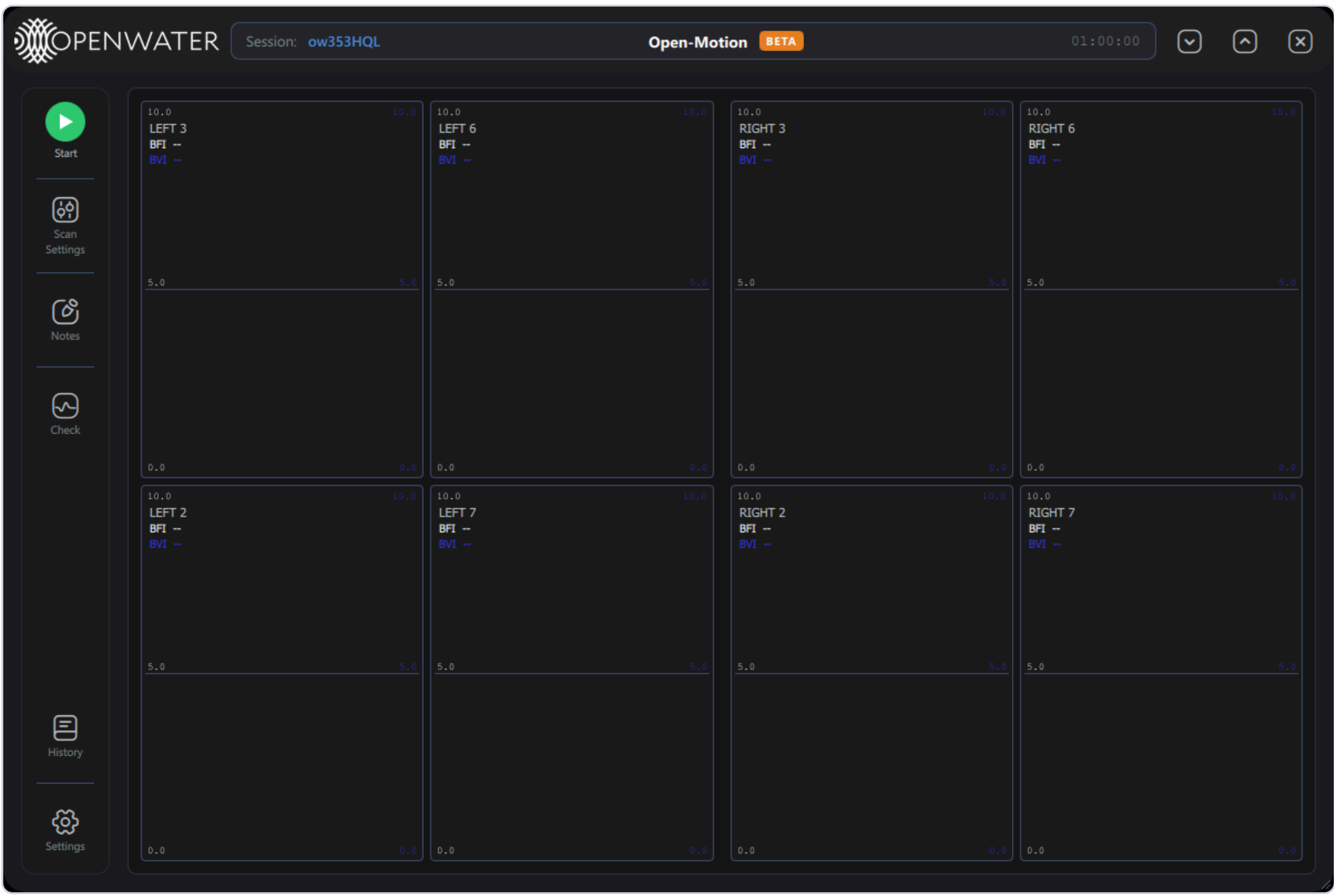
3. The main screen



#	Element	What it does
1	**Openwater logo**	Branding.
2	**Session:**	Current session identifier — regenerated per launch, or derived from the User Label you set in Scan Settings. Stamped into every scan record.
3	**BETA badge**	Marks Open-Motion Research.
4	**Scan clock**	Idle: the configured scan duration (e.g. `01:00:00`), or "Continuous". Scanning: elapsed / total, in green.
5	**Window controls**	Minimize `⌄`, maximize/restore `^`, close `✕`.
6	**Start / Stop badge**	Starts or stops the scan. Grey = disconnected, green = ready, yellow = start pending, red = scanning.
7	**Scan Settings**	Session label, camera patterns, and scan duration (§4). Disabled during a scan.

#	Element	What it does
8	**Notes**	Session Notes editor (§7).
9	**Check**	Runs an on-demand contact-quality check (§5).
10	**History**	Scan History: review, replay, export, delete (§8).
11	**Settings**	Application settings (§10).
12	**Plot grid**	One live cell per active camera, labeled `LEFT n` / `RIGHT n`.

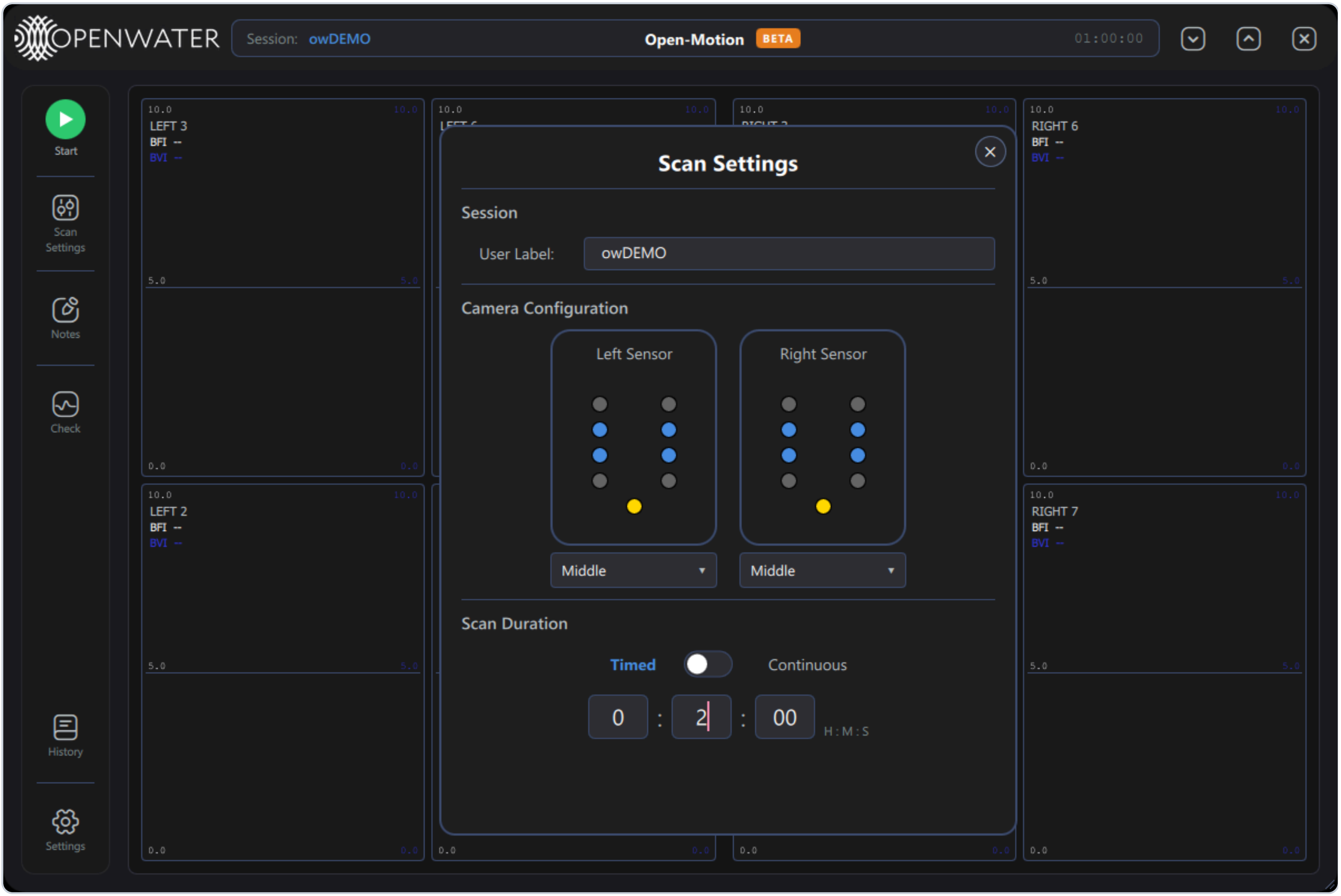
Window behaviors: drag the header to move; resize with the bottom-right grip (min 800 × 600); pressing `×` during a scan warns first — press it again within 5 s to cancel the work and exit.



Ready with the default Middle pattern: cameras 2, 3, 6, 7 on each sensor, one plot cell each.

4. Scan Settings

Press **Scan Settings** to configure the next scan:



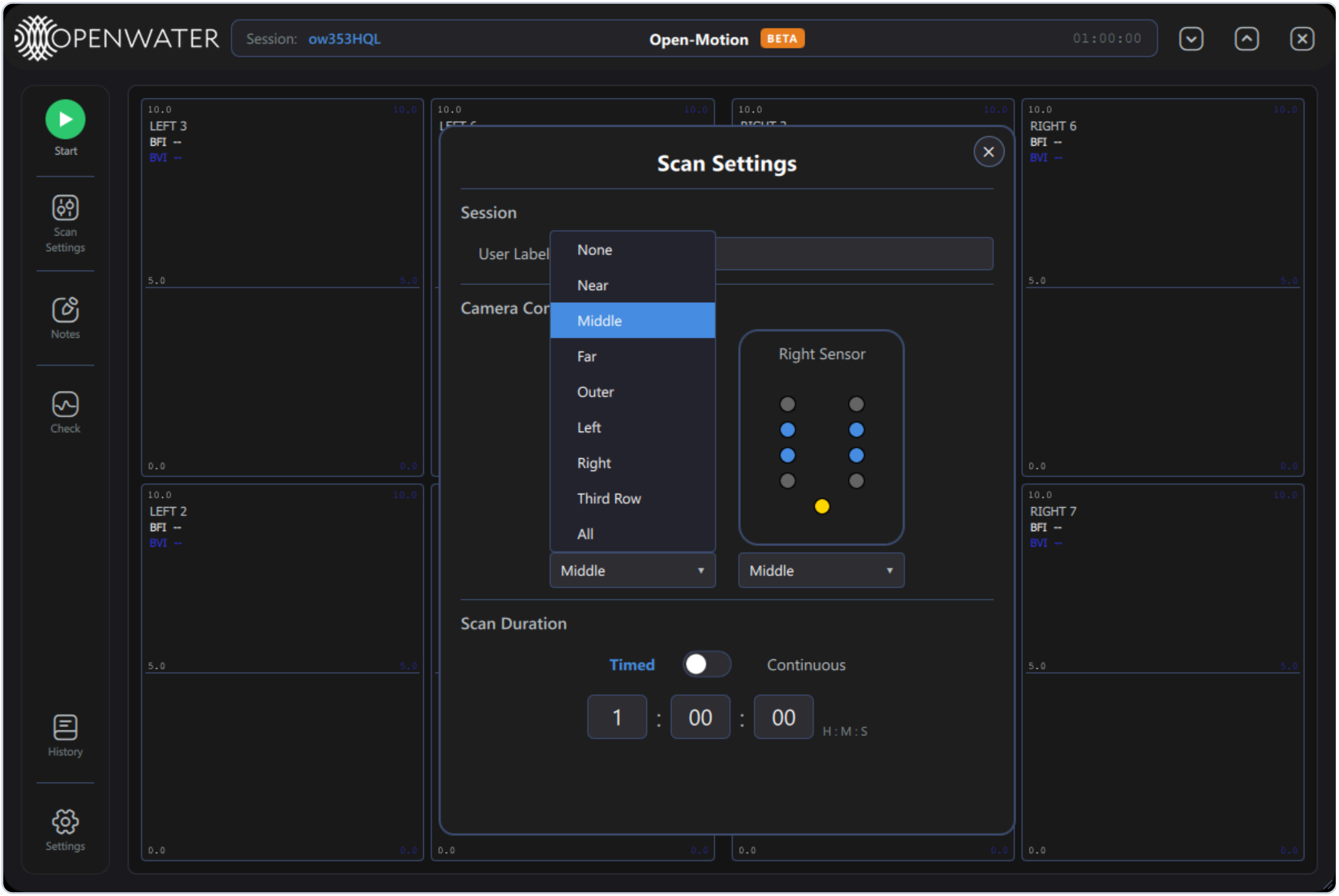
Session

Control	What it does
User Label	Free-text label for the session (committed when you leave the field). It becomes part of the scan's full label — e.g. label `DEMO` produces scans labeled `owDEMO` with full names like `20260718_051539_owDEMO`.

Camera Configuration

Each sensor card shows its 8 cameras (blue = active in the selected pattern, grey = inactive; the gold dot is the laser aperture). Hovering a dot shows the camera number. A disconnected sensor's card dims and its picker disables.

The dropdown under each card selects the camera pattern:



Pattern	Cameras enabled
None	none (side disabled)
Near	2, 4, 5, 7 — closest to the laser
Middle	2, 3, 6, 7 (default)
Far	1, 2, 7, 8 — farthest ring
Outer	1, 4, 5, 8
Left	1–4
Right	5–8
Third Row	2, 7
All	all 8 cameras

More active cameras = more coverage but denser plots and larger scan records.

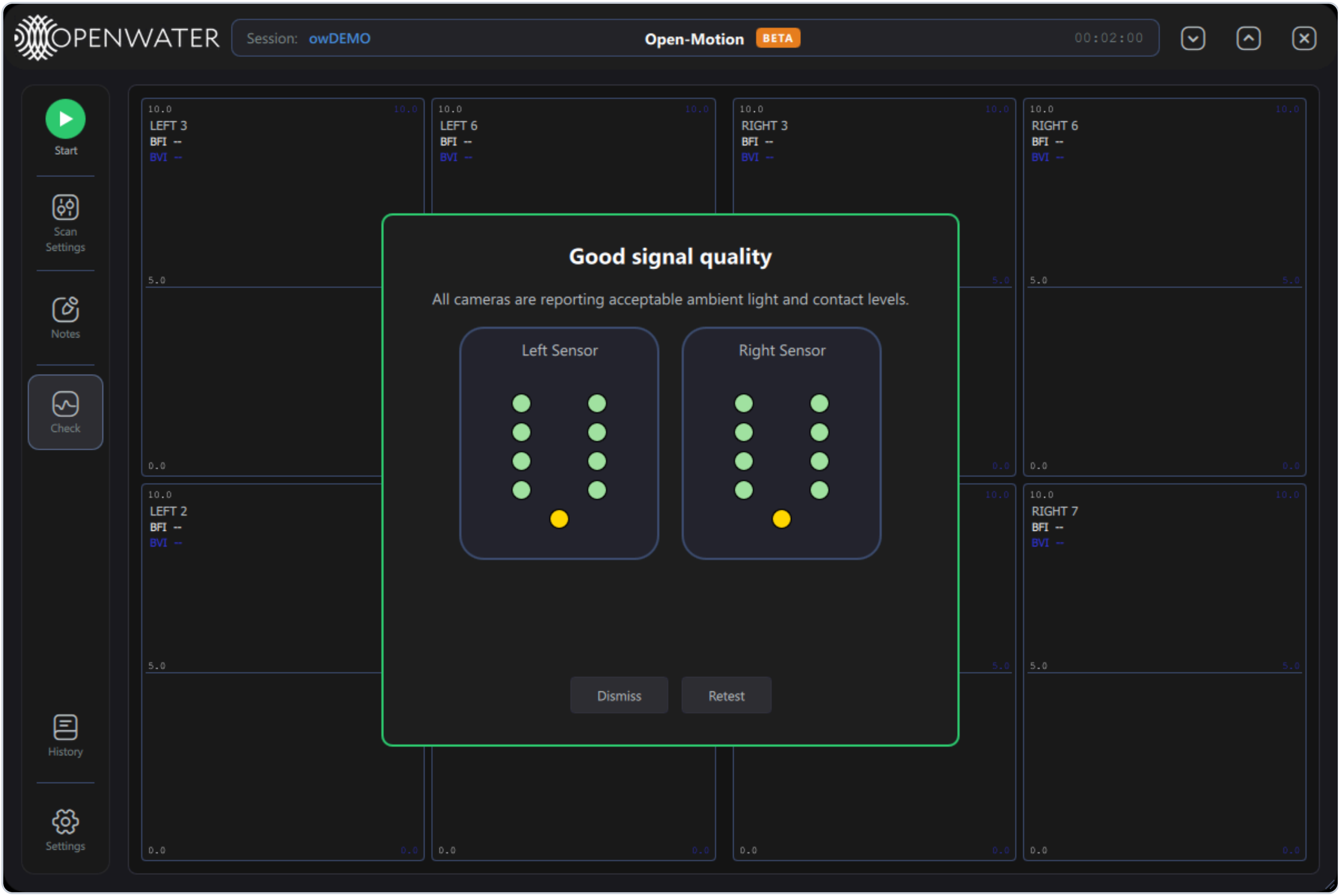
Scan Duration

Control	What it does
Timed / Continuous switch	*Timed*: the scan stops itself after the set duration. *Continuous*: runs until you press Stop ("Scan will run indefinitely until stopped.").
H : M : S fields	The timed duration (default 1:00:00). A duration of 0:00:00 is rejected — it resets to 1 minute with a warning toast.

Closing the dialog (✕) applies the selection. Your camera selection is kept even if a sensor briefly disconnects and reconnects.

5. Contact-quality Check

Press **Check** at any time (not during a scan) to measure ambient light and skin contact on every camera without recording data. The check briefly configures the cameras, samples for a moment, then reports:



- Panel border and title summarize the outcome — green "*Good signal quality*", orange "*Contact check failed*" (hover the orange dots for the specific reason), or red if the check could not run.
- Dot colors: **green** good · **grey** not evaluated · **orange** problem.
- **Dismiss** closes; **Retest** runs it again after you adjust the sensors.

During a live scan the same dialog appears automatically if contact quality degrades, with **Stop scan** and **Continue** buttons (Continue unlocks once the problem has stayed clear for a couple of seconds).

6. Running a scan

1. Configure Scan Settings (optional — defaults are sensible).
2. Optionally run **Check** and re-seat sensors until green.
3. Press **Start**. The scan launches directly — the badge turns yellow while the cameras are configured (typically a few seconds; up to ~1 min after a cold connect), then red as data starts flowing.
4. Watch the live grid:

Each cell shows one camera: `SIDE n`, live **BFI** and **BVI** numbers, the BFI trace (white) and BVI trace (blue). The header clock counts `elapsed / total`. The hover readout (top-right in the screenshot) lists every camera's values at the cursor time.

5. Press **Stop** (or let a timed scan complete). After teardown, **Session Notes** opens automatically with the stop event and duration pre-filled; add observations and close.

Scan data is stored in the local database automatically; no export is needed for retention.

Mid-scan annotations

Press **Space** during a scan to open Notes with a timestamp line pre-inserted:

The `[elapsed / wall-clock]` prefix is inserted for you — just type the observation.

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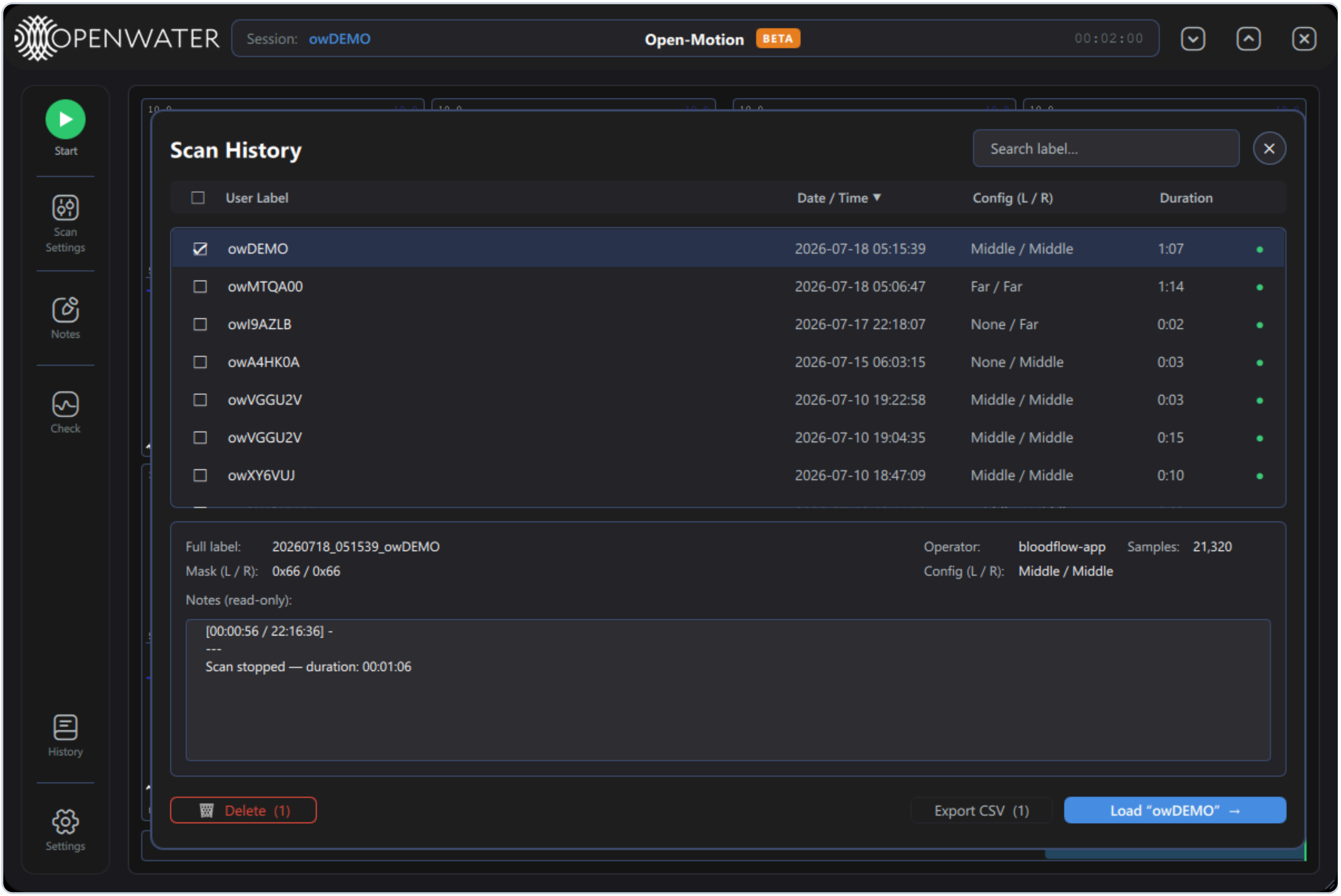
7. Session Notes

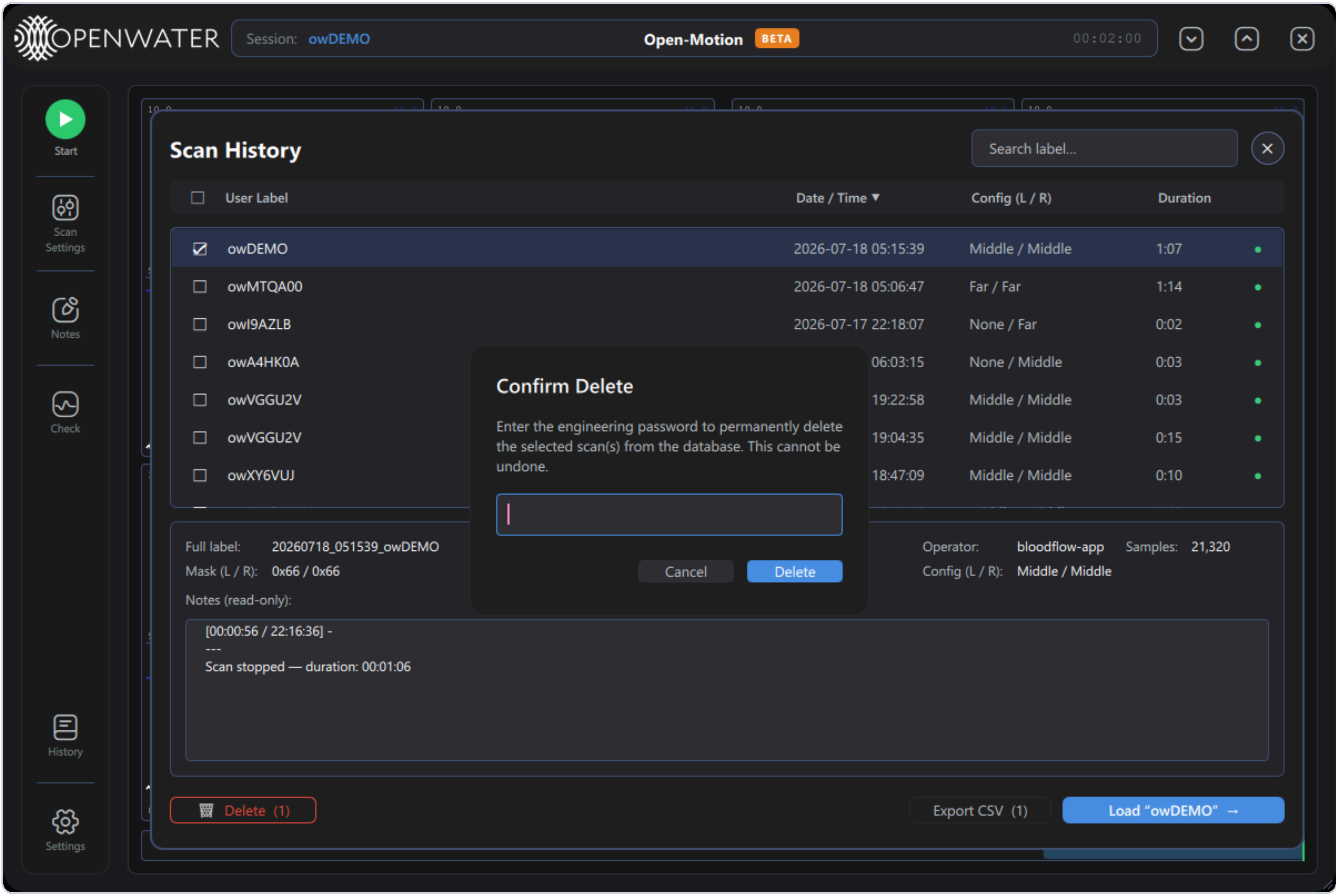
- Open via **Notes**, automatically after each scan, or with **Space** mid-scan.
- Text is saved to the current session's scan record when the window closes ("Note saved." toast).
- Notes appear read-only in the History detail pane.

8. Scan History



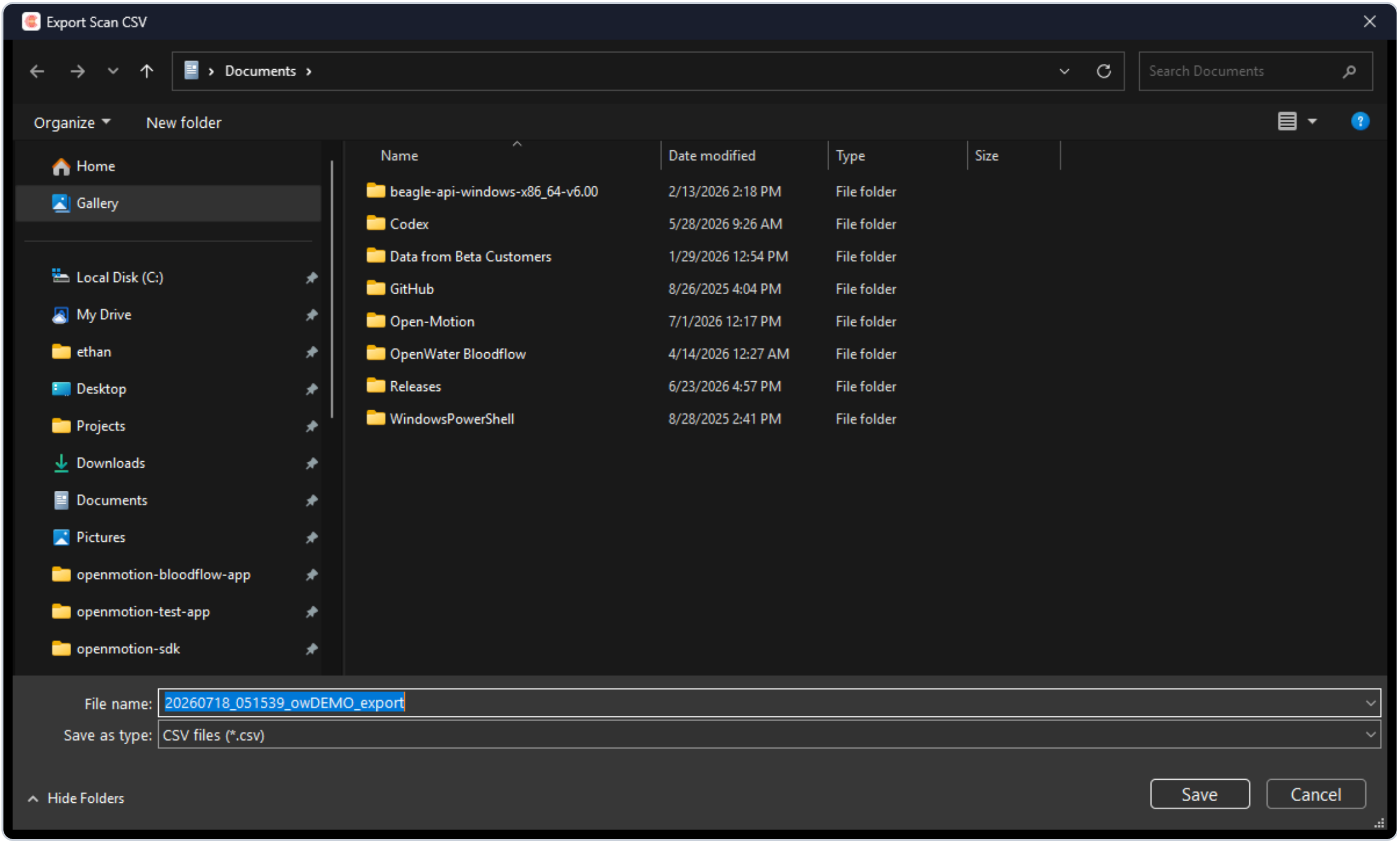
The list shows every scan in the database. Times in the **Date / Time** column are shown in UTC.

Control	What it does
Search label...	Filter by label text.
Column headers	Sort by User Label / Date / Config / Duration (▲/▼ toggles direction).
Row click	Select and show the detail pane: full label, operator, sample count, camera masks (hex), configuration names, and read-only notes.
Checkboxes	Mark rows for Delete / Export; the header checkbox selects all.
Status dot	Green ● complete · amber ⚠ interrupted (cannot be replayed or exported).
🗑 Delete (N)	Permanently deletes the checked scans — password-protected:



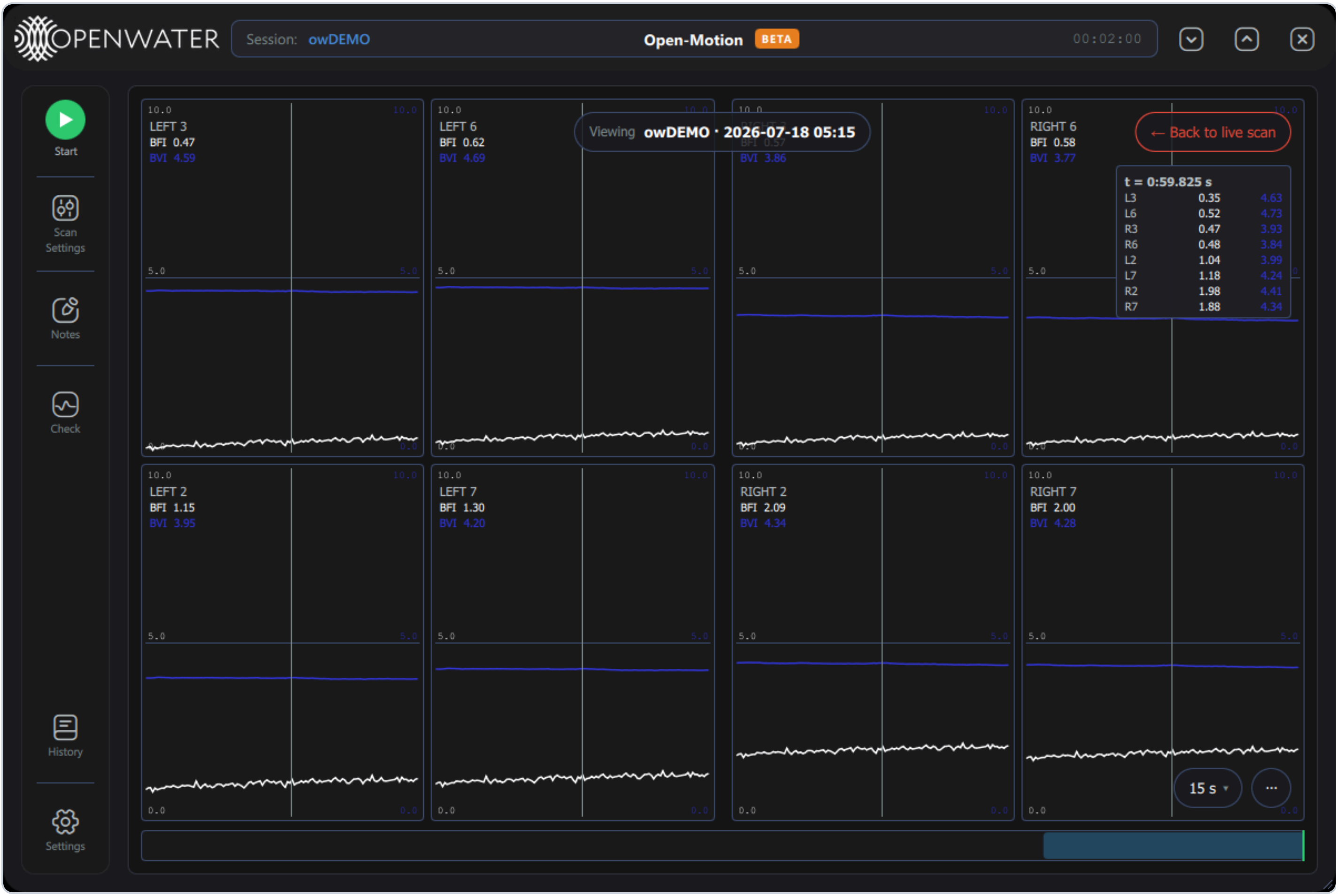
"Confirm Delete" — deletion requires a password and cannot be undone.

Control	What it does
Export CSV (N)	One checked scan → a save-file dialog (pre-named `⟨full label⟩_export.csv`). Several → a folder picker, one CSV per scan. Interrupted scans are skipped with a warning.



Control	What it does
Load "label" →	Loads the selected scan into the viewer for replay (a "Loading scan..." overlay appears for large scans).

Replay



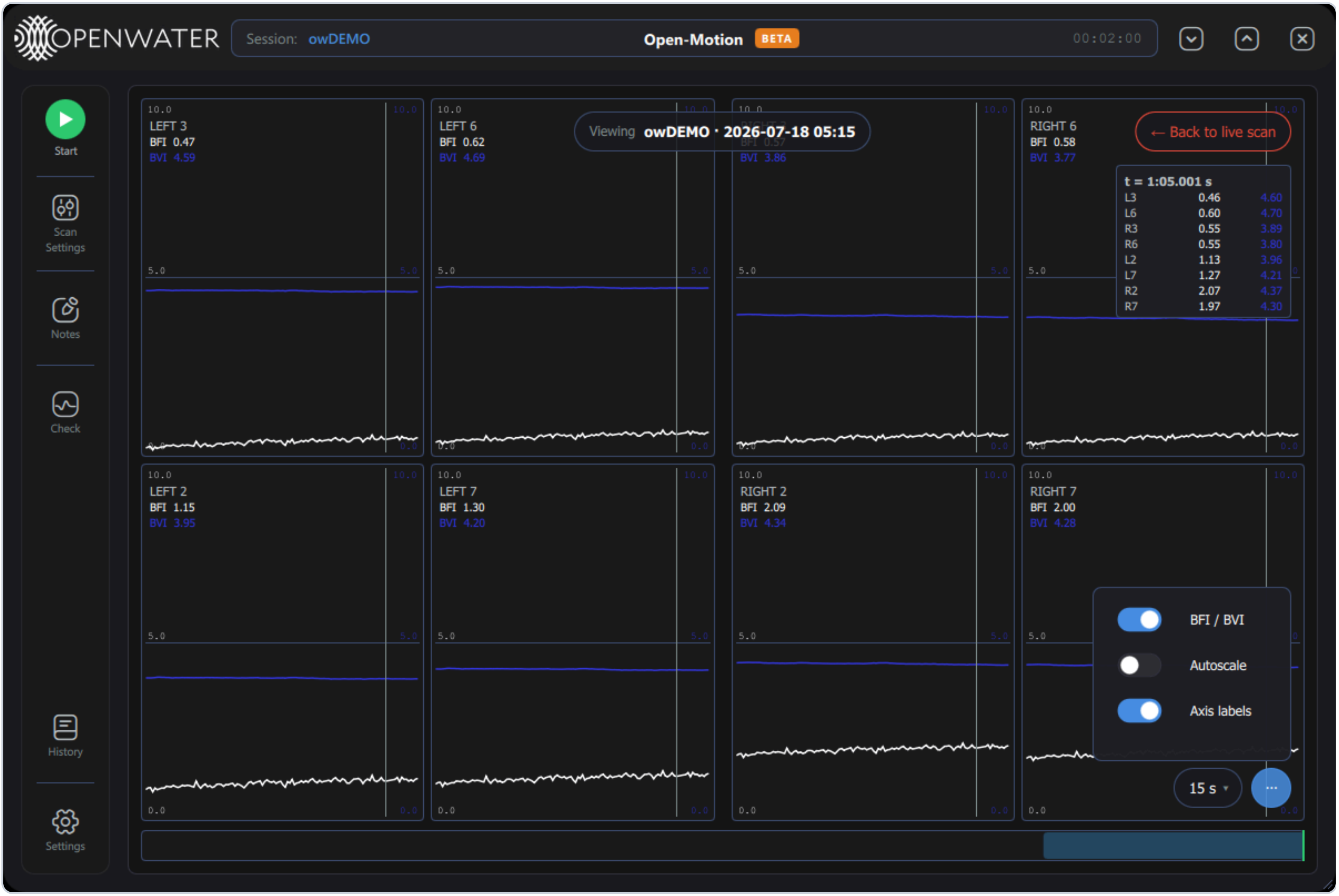
While replaying: the "*Viewing 〈label · date〉*" badge names the scan, the red **"← Back to live scan"** pill returns to the live view, and every navigation feature (timeline, zoom, hover, keyboard) works as in live view.

9. Plot viewer reference

9.1 Time navigation (DVR)

Control	What it does
`15 s ▾` **pill** (bottom-right)	Visible time window: 5 s / 15 s / 30 s / 1 min / 5 min.
Timeline bar (bottom)	The whole scan; drag the inset window to pan, click to jump. Blue inset = following live, orange = paused; a green tick marks the live edge.
Drag on a plot	Pan in time (pauses live-follow).
Wheel on a plot	Zoom the time window around the cursor (pinch works on touch screens).
Hover	Synchronized crosshair on all cells + per-camera values at the cursor time.
`● Back to live`	Return to the live edge after panning into the past.

9.2 Display options — the `···` menu



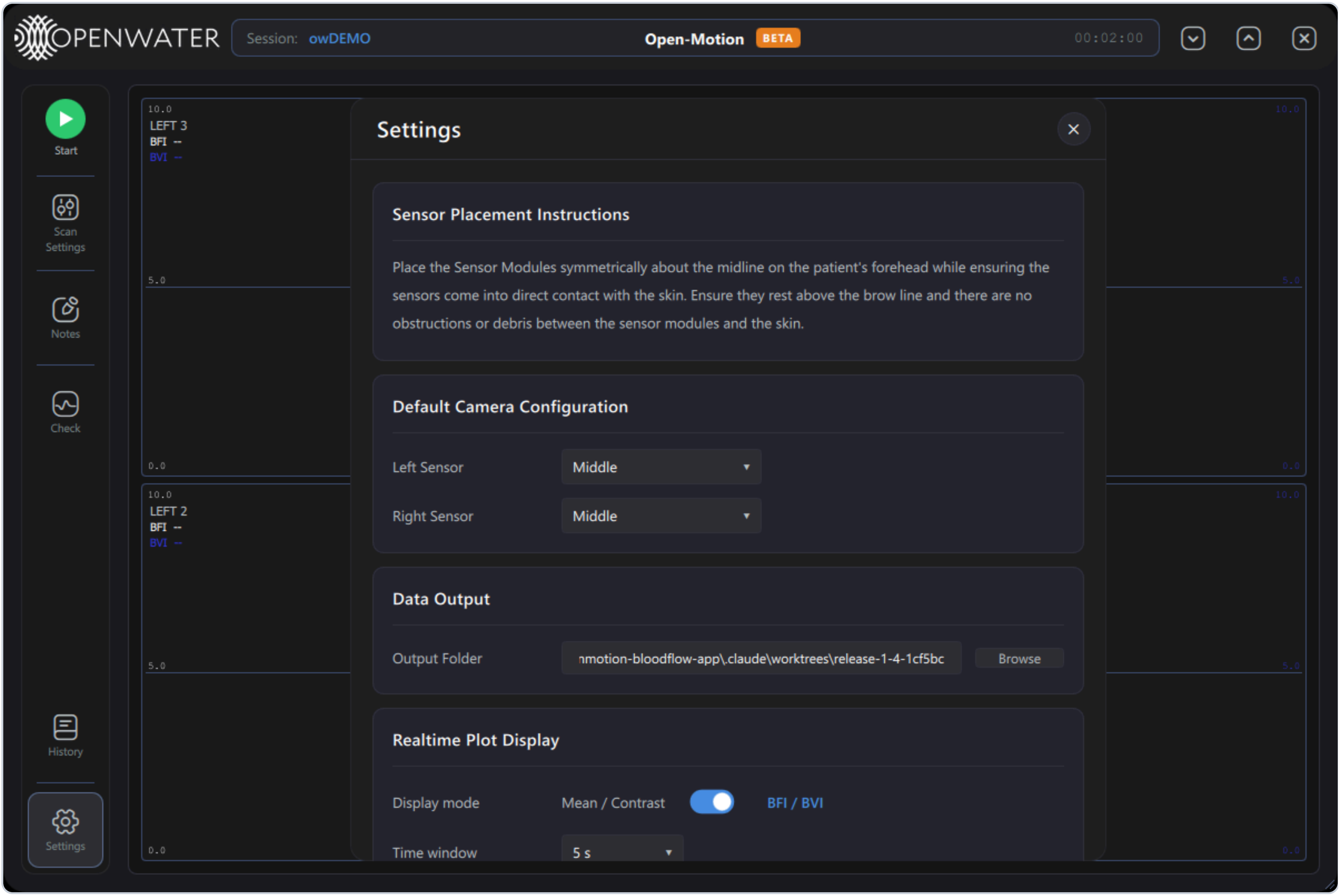
Toggle	What it does
BFI / BVI	Switches every cell between BFI/BVI and Mean/Contrast (the raw speckle statistics BFI/BVI are derived from).
Autoscale	Automatic Y-axis ranging per metric (recomputed every few seconds). Off = the fixed Manual Plot Bounds from Settings.
Axis labels	Shows/hides the numeric Y-axis tick labels on each cell.

9.3 Keyboard shortcuts

Key	Action
`←` / `→`	Pan 1 s (`Shift` = one full window)
`+` / `↑`	Zoom in
`-` / `↓`	Zoom out
`0`	Reset window and return to live

Key	Action
`Home` / `End`	Scan start / live edge
`Space`	Timestamped note (during a scan)
`Esc`	Close the current dialog
`Enter`	Confirm in password prompts

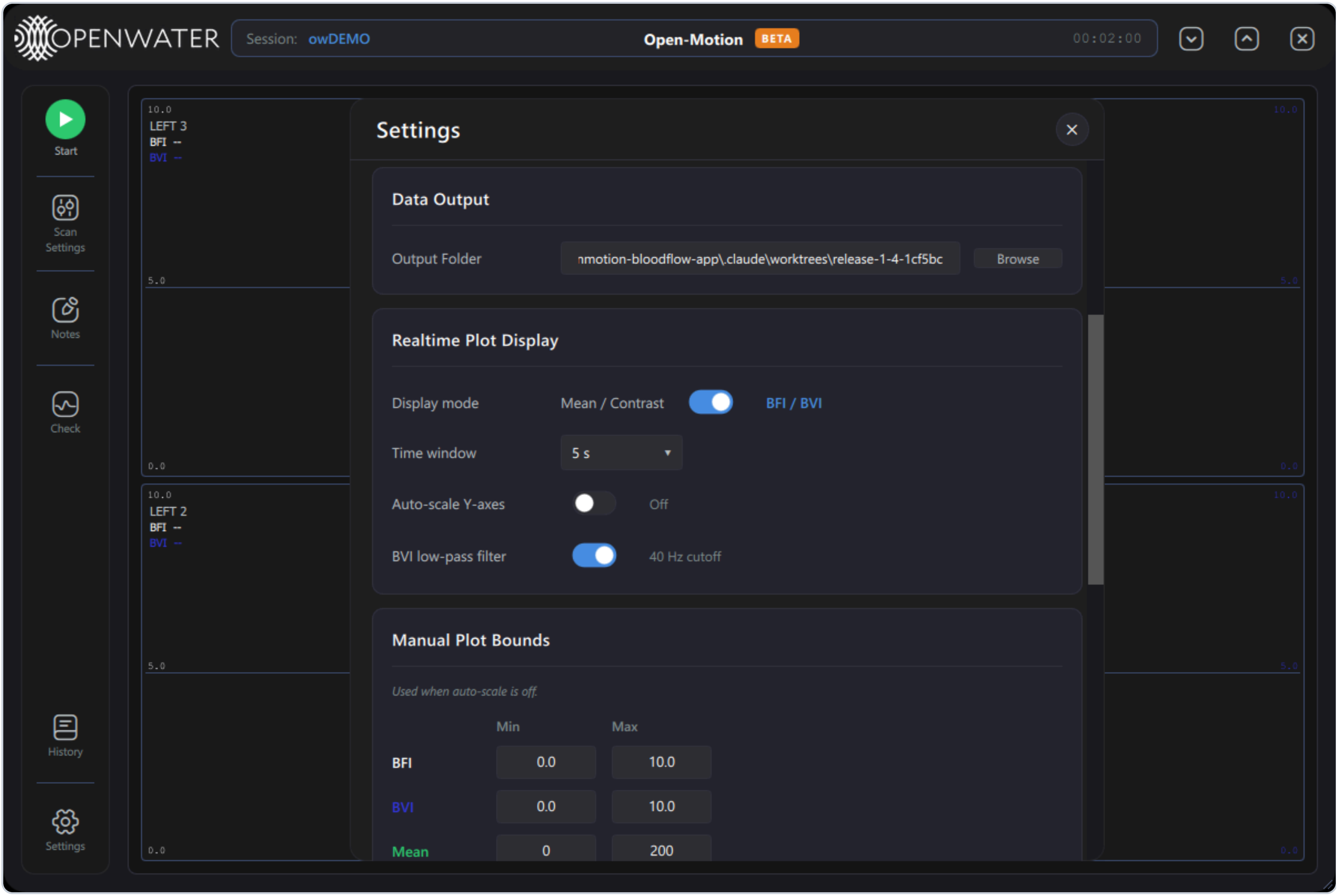
10. Settings



Sensor Placement Instructions — placement reference text.

Default Camera Configuration — the default *Left Sensor / Right Sensor* patterns pre-selected at launch (same nine patterns as Scan Settings). Scan Settings overrides them for the current session.

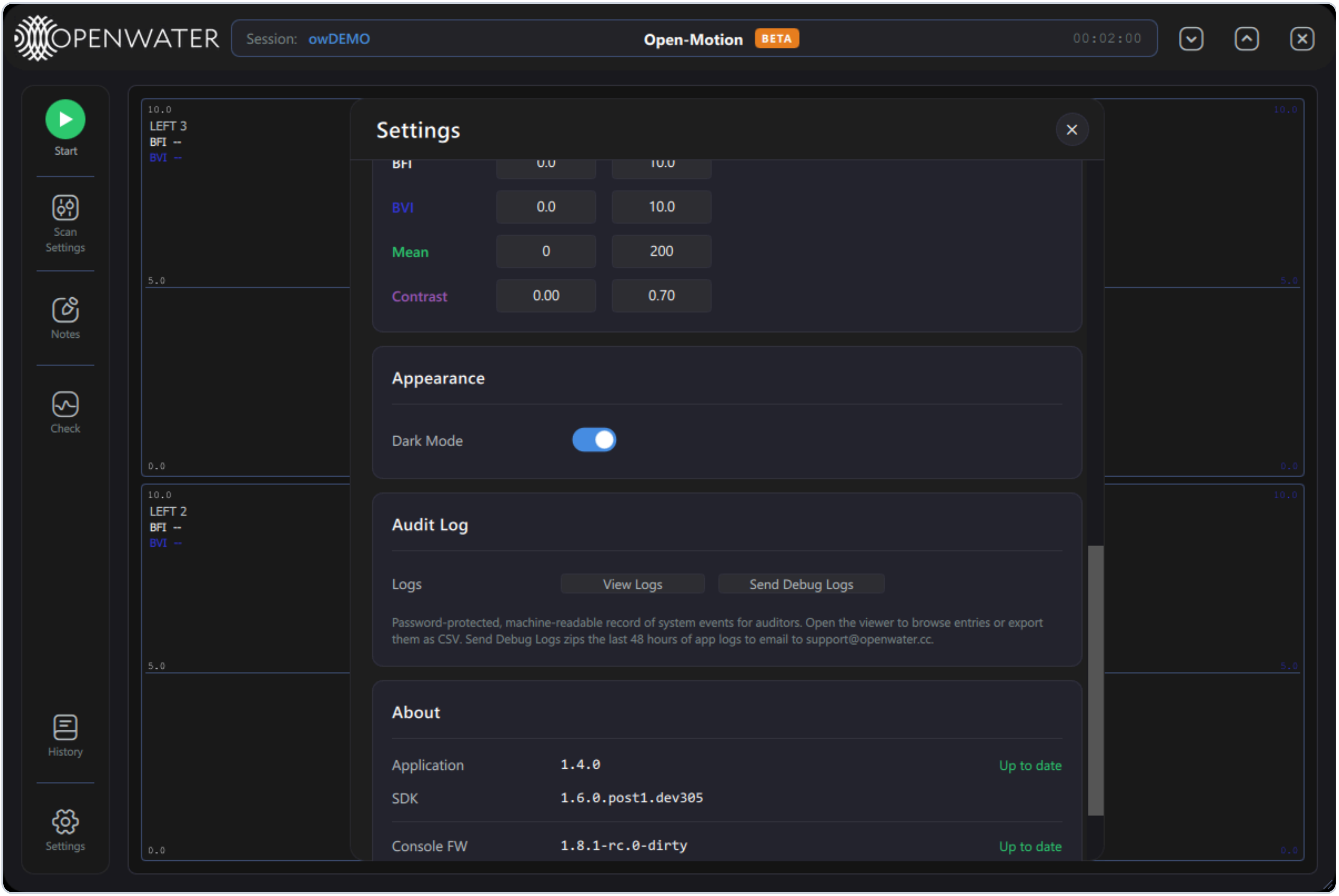
Data Output — *Output Folder* (where the database, logs and exports live) with **Browse**.



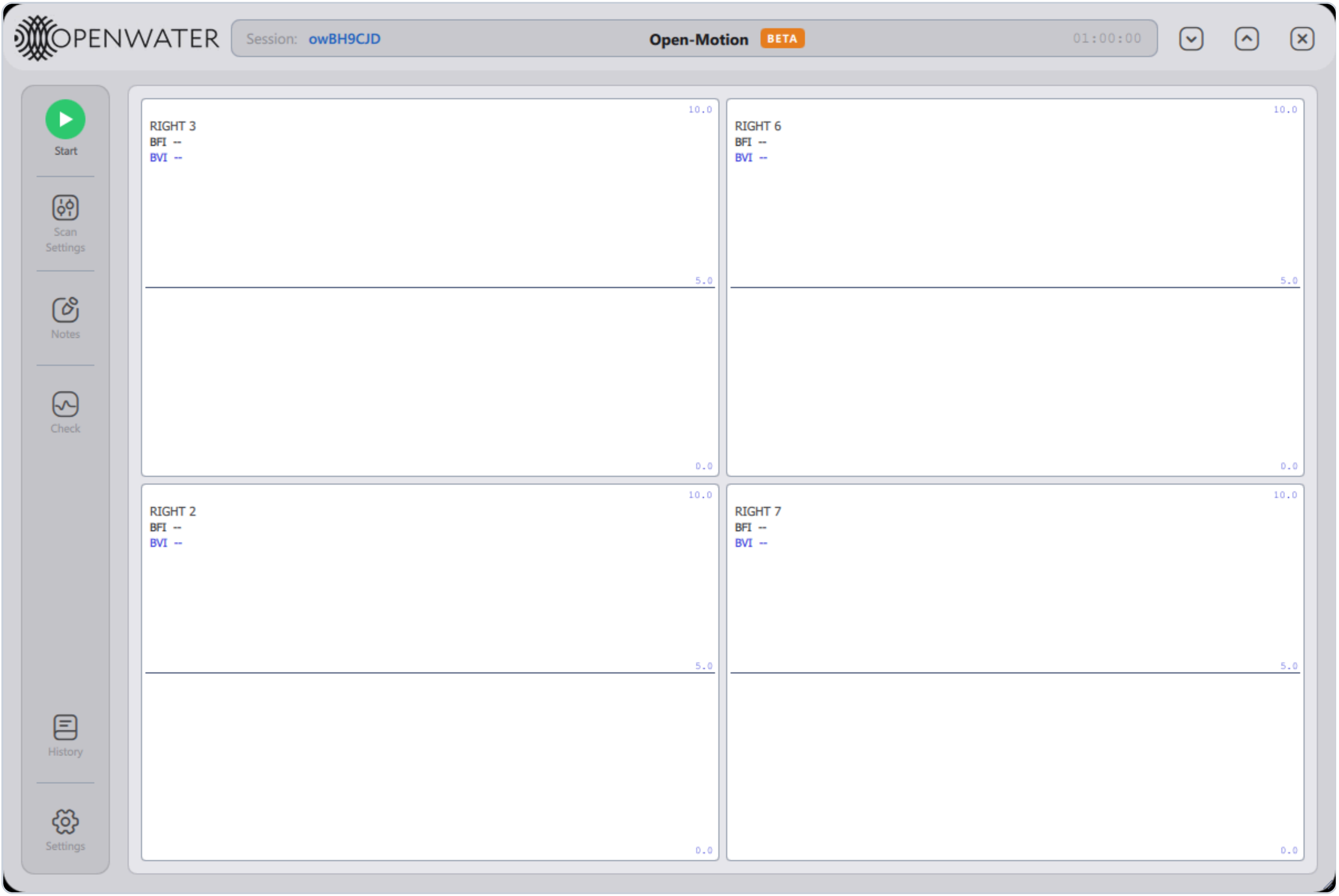
Realtime Plot Display

Setting	What it does
Display mode	Mean / Contrast ↔ BFI / BVI (same as the plot `⋯` menu).
Time window	Default visible window length.
Auto-scale Y-axes	Same as the plot `⋯` menu's Autoscale.
BVI low-pass filter	Smooths the BVI trace (40 Hz cutoff).

Manual Plot Bounds — fixed Y ranges used when auto-scale is off: Min/Max for BFI, BVI, Mean and Contrast.



Appearance — *Dark Mode* switch; the theme flips immediately:



The light theme. (Shown here with only the right sensor connected — the grid always shows exactly the connected, active cameras.)

Audit Log — **View Logs** (password-protected audit-log viewer for auditors and Openwater support) and **Send Debug Logs** (zips the last 48 h of logs for emailing to **support@openwater.cc**).

About — Application / SDK / Console FW / Left & Right Sensor FW versions with up-to-date status. When a newer application release exists, an **Update** chip appears next to the outdated component.

Closing Settings saves any changes.

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11. Alerts and errors

Toasts appear bottom-right (green success, yellow warning, red error, blue info; hover pauses auto-dismiss; safety-critical ones persist). Blocking failures raise the **Critical Error** dialog with an error code, suggested action, **Copy details**, **Send Bug Report to Openwater**, an expandable **Details** block, and **Dismiss**.

Code	Title	Suggested action (summary)
E-101	Sensor self-check failed	Power-cycle the sensor; service if persistent.
E-102	Sensor self-check unreadable	Power-cycle; update firmware or report.
E-103	Console initialization failed	Power-cycle the console; report if persistent.
E-105	Camera power-on failed	Power-cycle the sensor; possible service.
E-201	Laser safety monitor unresponsive	Power-cycle; do not scan until clear.
E-202	Laser safety trip	Clear obstruction, settle, rescan.
E-301	Scan aborted before start	Resolve the precondition and retry.
E-302	Scan could not start	Wait for the previous scan to finish.
E-303	Camera data lost during scan	Check cables/power; data before the loss was saved.

(Full error wording is listed in the *Open-Motion User Manual*, §9.3; it is identical in both applications.)

12. Data storage

Location (under the Output Folder)	Contents
`logs/`	One application log per launch — first stop when troubleshooting.
`data/scans.db`	Scan database: all sessions, per-camera data, notes.
`data/debug-bundles/`	Send-Debug-Logs zips.

CSVs are export-time artifacts only (History → Export CSV).